Chapter 4: Quantum teleportation

Gyuyoung Hwang

April 15, 2022

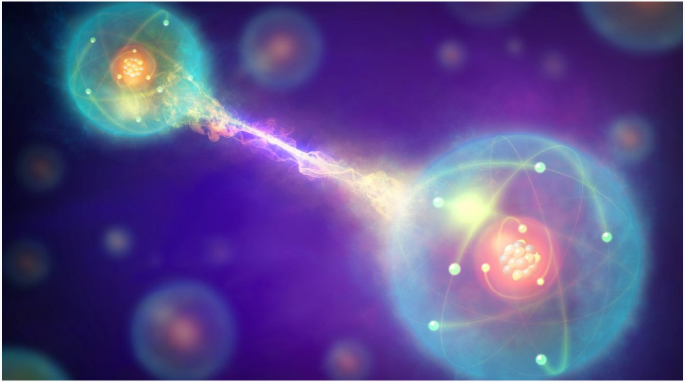
Seoul National University
hgy0407@snu.ac.kr

Table of Contents

Table of Contents

Quantum Entanglement : A review

"When two or more particles link up in a certain way, no matter how far apart they are in space, their states remain linked.
That means they share a common, unified quantum state.
So observations of one of the particles can automatically provide information about the other entangled particles, regardless of the distance between them." - Paul Sutter, 'What is quantum entanglement?'.



Definition

A state $\rho \in D(\mathbb{H}^A \otimes \mathbb{H}^B)$ in a composite system $\mathbb{H}^A \otimes \mathbb{H}^B$, which is composed of the sub-systems $\mathbb{H}^A$ and $\mathbb{H}^B$ is called separable with respected to the sub-systems $\mathbb{H}^A$ and $\mathbb{H}^B$, if there exist states $\rho_j^A \in D(\mathbb{H}^A)$ in sub-system $\mathbb{H}^A$ and $\rho_j^B \in D(\mathbb{H}^B)$ in sub-system $\mathbb{H}^B$ indexed by $j \in I \subset \mathbb{N}$ together with positive real numbers p_j satisfying

$$\sum_{j \in I} p_j = 1,$$

such that

$$\rho = \sum_{j \in I} p_j \rho_j^A \otimes \rho_j^B .$$

Otherwise, ρ is called entangled.

For a pure state $|\psi\rangle \in \mathbb{H}^A \otimes \mathbb{H}^B$, view this as a mixed state by writing

$$\rho = |\psi\rangle\langle\psi|.$$

Then we see that it is separable if and only if

$$|\psi\rangle = |\phi\rangle \otimes |\theta\rangle, \quad \text{for some} \quad |\phi\rangle \in \mathbb{H}^A \quad \text{and} \quad |\theta\rangle \in \mathbb{H}^B.$$

Quantum Entanglement : A review

Example

A Bell state $|\Phi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$ is an entangled state. Indeed, they are maximally entangled.

Example

A GHZ state $|GHZ\rangle = \frac{|0\rangle^{\otimes M} + |1\rangle^{\otimes M}}{\sqrt{2}}$.

Definition

A pure state $|\Psi\rangle$ in the tensor product of identical HIlbert spaces $\mathbb{H}^A$ is said to be maximally entangled if

$$\rho^A(\Psi) = \lambda 1, \quad \text{with} \quad 0 < \lambda < 1.$$

Meanwhile, for an entangled state, i.e an element in $\mathcal{H}_1 \bigotimes \mathcal{H}_2$, we want to reduce the system into one Hilbert space $\mathcal{H}_1$. To this end, the partial trace is defined by the unique linear operator

$$Tr_{\mathcal{H}_2} : L(\mathcal{H}_1 \bigotimes \mathcal{H}_2) \to L(\mathcal{H}_1), Tr_{\mathcal{H}_2}(R \bigotimes S) = Tr(S)R.$$

Table of Contents

Definition

Given (1) an arbitrary (original) state $|\psi\rangle \in \mathbb{H}$ to be copied and (2) a (white-page) state $|\omega\rangle \in \mathbb{H}$ tp emerge as a copy, a quantum-copier K is a linear transformation that leaves the original state $|\psi\rangle$ unchanged and transforms $|\omega\rangle$ such that it becomes $|\psi\rangle$, i.e., K is an operator satisfying

$$K : |\psi\rangle \otimes |\omega\rangle \mapsto |\psi\rangle \otimes |\psi\rangle$$

for arbitrary $|\psi\rangle \in \mathbb{H}$ and a given fixed $|\omega\rangle \in \mathbb{H}$.

Theorem (Quantum No-Cloning Theorem)

A quantum-copier cannot exist.

Proof of No-Cloning Theorem

We must have

$$K(|0\rangle \otimes |\omega\rangle) = |0\rangle \otimes |0\rangle, \quad K(|1\rangle \otimes |\omega\rangle) = |1\rangle \otimes |1\rangle,$$

and also,

$$K\left(\frac{|1\rangle + |0\rangle}{\sqrt{2}} \otimes |\omega\rangle\right) = \frac{|1\rangle + |0\rangle}{\sqrt{2}} \otimes \frac{|1\rangle + |0\rangle}{\sqrt{2}}.$$

However, we also have

$$K\left(\frac{|1\rangle + |0\rangle}{\sqrt{2}} \otimes |\omega\rangle\right) = \frac{1}{\sqrt{2}}(|1\rangle \otimes |1\rangle + |0\rangle \otimes |0\rangle).$$

Quantum Teleportation

Dense Quantum Coding : Transmit two classical bits by only sending one qubit

Quantum Teleportation : Transmit one qubit by sending two classical bits.

Basic Theme : Use entanglement, in particular use Bell state.

Bell states and measurement : A review

Notation :

1. Bell basis

$$
\begin{aligned}
|\Phi^{\pm}\rangle &:= \frac{1}{\sqrt{2}}\Big(|00\rangle \pm |11\rangle\Big) \\
|\Psi^{\pm}\rangle &:= \frac{1}{\sqrt{2}}\Big(|01\rangle \pm |10\rangle\Big).
\end{aligned}
$$

2. Paul matrices

$$
\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}
$$

Bell states and measurement : A review

3. By direct computation, one has

$$
\begin{aligned}
(\sigma_z \otimes \sigma_z)|\Phi^{\pm}\rangle &= |\Phi^{\pm}\rangle \\
(\sigma_z \otimes \sigma_z)|\Psi^{\pm}\rangle &= -|\Psi^{\pm}\rangle \\
(\sigma_x \otimes \sigma_x)|\Phi^{\pm}\rangle &= \pm|\Phi^{\pm}\rangle \\
(\sigma_x \otimes \sigma_x)|\Psi^{\pm}\rangle &= \pm|\Psi^{\pm}\rangle.
\end{aligned}
$$

Therefore, we can distinguish each Bell basis vectors by measuring with respect to the operators $\sigma_z \otimes \sigma_z$ and $\sigma_x \otimes \sigma_x$.

Dense quantum coding

We start with dense quantum coding.
Alice wants to send two classical bits x_1, x_0 to Bob.

1. Prepare a Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.
2. Apply a quantum gate $U^A(x_1, x_0)$ to $|\Phi^+\rangle$.
3. Measures $\sigma_z^A \otimes \sigma_z^B$ and $\sigma_x^A \otimes \sigma_x^B$ and observes two values (one of $1, -1$ each).

Dense quantum coding

Table 6.1 Protocol of dense quantum coding

Alice wants to send the classical bits x_1x_0	So she applies $U^A(x_1x_0)$	The state of the total system becomes $\left(U^A \otimes \mathbf{1}\right)\lvert\Phi^+\rangle$	on which Bob measures $\sigma_z^A \otimes \sigma_z^B$ and $\sigma_x^A \otimes \sigma_x^B$ and observes the values
00	$\mathbf{1}^A$	$\lvert\Phi^+\rangle$	$+1\,,+1$
01	σ_z^A	$\lvert\Phi^-\rangle$	$+1\,,-1$
10	σ_x^A	$\lvert\Psi^+\rangle$	$-1\,,+1$
11	$\sigma_z^A\sigma_x^A$	$\lvert\Psi^-\rangle$	$-1\,,-1$

$$
\begin{aligned}
U^A(00) &= \mathbf{1}^A \\
U^A(01) &= \sigma_z^A \\
U^A(10) &= \sigma_x^A \\
U^A(11) &= \sigma_z^A\sigma_x^A\,.
\end{aligned}
\tag{6.6}
$$

Quantum Teleportation : Theory

1. Alice wants to teleport the qubit $|\psi\rangle = a|0\rangle + b|1\rangle \in \mathbb{H}^S$ to Bob.
2. For an entangled Bell state,

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|0\rangle^A \otimes |0\rangle^B + |1\rangle^A \otimes 1\rangle^B) \in \mathbb{H}^A \otimes \mathbb{H}^B,$$

suppose part of it belongs to Alice and Bob has the remainder.

Quantum Teleportation : Theory

3. Alice then applies $\sigma_z^S \otimes \sigma_z^A$ and $\sigma_x^S \otimes \sigma_x^A$ to her qubit in $\mathbb{H}^S \otimes \mathbb{H}^A$.

4. According to the result of measurement, the three-qubit total state is determined.

5. Bob determines the operator to apply and get the qubit $|\psi\rangle$.

Quantum Teleportation : Theory

Important computation

$$
\begin{aligned}
|\psi\rangle^S \otimes |\Psi^+\rangle^{AB} &= \Big(a|0\rangle^S + b|1\rangle^S\Big) \otimes \frac{1}{\sqrt{2}}\Big(|0\rangle^A \otimes |0\rangle^B + |1\rangle^A \otimes |1\rangle^B\Big) \\
&= \frac{1}{2}\Big[|\Psi^+\rangle^{SA} \otimes |\psi\rangle^B + |\Psi^+\rangle^{SA} \otimes \Big(\sigma_x^B|\psi\rangle^B\Big) \\
&\quad + |\Psi^-\rangle^{SA} \otimes \Big(\sigma_z^B|\psi\rangle^B\Big) + |\Psi^-\rangle^{SA} \otimes \Big(\sigma_x^B\sigma_z^B|\psi\rangle^B\Big),
\end{aligned}
$$

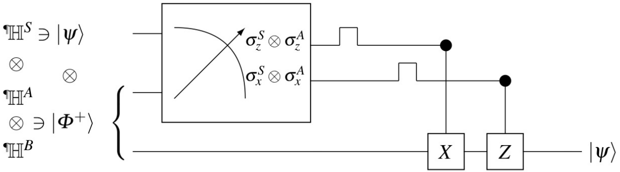
where we used

$$
|0\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}}\Big(|\Psi^+\rangle^{SA} + |\Psi^-\rangle^{SA}\Big), \cdots .
$$

Quantum Teleportation : Theory

Table 6.2 Protocol for teleportation

Alice measures $\sigma_z^S \otimes \sigma_z^A$ and $\sigma_x^S \otimes \sigma_x^A$ to observe	The three-qubit total state after measurement is then: $\|\Psi\rangle^{SA} \otimes$ Bob's qubit-state	From bits received Bob determines to apply U^B	The state of Bob's qubit then becomes
$+1\,,+1$	$\|\Phi^+\rangle \otimes \|\psi\rangle$	$\mathbf{1}^B$	$(\mathbf{1}^B)^2\|\psi\rangle = \|\psi\rangle$
$+1\,,-1$	$\|\Phi^-\rangle \otimes \sigma_z^B\|\psi\rangle$	σ_z^B	$(\sigma_z^B)^2\|\psi\rangle = \|\psi\rangle$
$-1\,,+1$	$\|\Psi^+\rangle \otimes \sigma_x^B\|\psi\rangle$	σ_x^B	$(\sigma_x^B)^2\|\psi\rangle = \|\psi\rangle$
$-1\,,-1$	$\|\Psi^-\rangle \otimes \sigma_x^B\sigma_z^B\|\psi\rangle$	$\sigma_z^B\sigma_x^B$	$\sigma_z^B\sigma_x^B\sigma_x^B\sigma_z^B\|\psi\rangle = \|\psi\rangle$



Quantum Teleportation : Implementation

```python
import cirq
import cirq_google
import numpy as np
```

```python
import random
# Create qubits : msg = qubit[0], qalice = qubit[1], qbob = qubit[2]
qubit=[0] *(3)
qubit[0] = cirq.NamedQubit('msg')
qubit[1] = cirq.NamedQubit('qalice')
qubit[2] = cirq.NamedQubit('qbob')
print(qubit)
circuit = cirq.Circuit()
```

```
[cirq.NamedQubit('msg'), cirq.NamedQubit('qalice'), cirq.NamedQubit('qbob')]
```

Quantum Teleportation : Implementation

```
#Create a Bell state entangled pair
circuit.append([cirq.H(qubit[1]), cirq.CNOT(qubit[1],qubit[2])])
print(circuit)
```

```
qalice: ───H───@───
               │
qbob:   ───────X───
```

Quantum Teleportation : Implementation

```python
#Create a random state for the message , repeating this may cause error
ranX = random.random()
ranY = random.random()
print(ranX, ranY)
circuit.append([cirq.X(qubit[0]) ** ranX, cirq.Y(qubit[0]) ** ranY])
print(circuit)
```

```
0.7511826460115856 0.7631230865776707
msg: ───────────X^0.751─────Y^0.763─────

qalice: ─────H────────────────@───────────────
                    │
qbob: ─────────────────────────X───────────────
```

Quantum Teleportation : Implementation

```python
#Unitary operator rotating the two-qubit basis of the message
#and Alice's entangled qubit
circuit.append([cirq.CNOT(qubit[0],qubit[1]), cirq.H(qubit[0])])
print(circuit)
```

```
msg: ──────────X^0.502────Y^0.233────@────H─────
                                     │
qalice: ───H───────────────@─────────X─────────────
           │
qbob: ─────────────────────X────────────────────────
```

```python
#Measurement
circuit.append(cirq.measure(qubit[0], qubit[1]))
print(circuit)
```

```
msg: ──────────X^0.502───Y^0.233───@───H───M───
                                   │   │
qalice: ───H───────────────────@───────X───────M───
                               │
qbob: ─────────────────────────X───────────────────
```

```
#Using two classical bits to recover the original message
circuit.append([cirq.CNOT(qubit[1],qubit[2]), cirq.CZ(qubit[0], qubit[2])])
print(circuit)
```

```
msg: ──────────X^0.502───Y^0.233───@───H───M──────────@──────
                               │       │     │
qalice: ───H────────────────@──────────X─────────M────@───┼──────
                   │                         │   │
qbob: ──────────────────────X─────────────────────────────X───@──────
```

Quantum Teleportation : Implementation

```
sim = cirq.Simulator()
q0 = cirq.LineQubit(0)
message = sim.simulate(cirq.Circuit([cirq.X(q0)**ranX, cirq.Y(q0)**ranY]))
print("\nBloch Sphere of the Message qubit in the initial state:")
expected = cirq.qis.bloch_vector_from_state_vector(message.final_state_vector, 0)
print(" x: ", np.around(expected[0], 4), " y:", np.around(expected[1],4), " z: ", np.around(expected[2],4))
```

```
Bloch Sphere of the Message qubit in the initial state:
 x:  -0.0036  y: -1.0  z:  -0.004
```

$\rightsquigarrow |x\rangle.$

Quantum Teleportation : Implementation

```
final_results = sim.simulate(circuit)
print("\nBloch Sphere of Bob's qubit in the final state:")
teleported = cirq.bloch_vector_from_state_vector(final_results.final_state_vector, 2)
print("x: ", np.around(teleported[0], 4), " y: ", np.around(teleported[1], 4), " z: ", np.around(teleported[2], 4))
print("\nBloch Sphere of the Message qubit in the final state:")
message_final = cirq.bloch_vector_from_state_vector(final_results.final_state_vector, 0)
print("x: ", np.around(message_final[0], 4), " y: ", np.around(message_final[1], 4), " z: ", np.around(message_final[2], 4))
```

```
Bloch Sphere of Bob's qubit in the final state:
x:  -0.0036  y:  -1.0  z:  -0.004

Bloch Sphere of the Message qubit in the final state:
x:  0.0  y:  0.0  z:  1.0
```

$|x\rangle$

Some comments on Quantum Teleportation

Interesting paper : Pirandola, S.; Eisert, J.; Weedbrook, C.; Furusawa, A.; Braunstein, S. L. (2015). "Advances in Quantum Teleportation". Nature Photonics

1. What we saw in previous slide is the qubit-teleportation ($d = 2$): Input is pure, entangled state is a Bell pair, and we apply a qubit Pauli operator.
2. For continuous-variable system ($d = \infty$) : The systems are bosonic modes, input is a coherent state, entangled state is an EPR state, and operator is a phase-space displacement.
3. Entanglement swapping : "Entanglement" itself can be teleported, i.e. the qubit to be transfered can be part of an entangled state.
4. Quantum gates teleportation : One can teleport quantum gates. Experimentally, two-qubit gates have been teleported.

Some comments on Quantum Teleportation

1. Issue of Fidelity : Teleportation fidelity should exceed an appropriate threshold.

   Given to density operators ρ and σ, the fidelity $F(\rho,\sigma)$ is defined by

   $$F(\rho,\sigma) = \left(Tr(\sqrt{\sqrt{\rho}\sigma\sqrt{\rho}} \right)^2,$$

   for the case of pure states, we have $F(\rho,\sigma) = |\langle\psi_\rho,\psi_\sigma\rangle|^2$.
   For example, we have $F(\rho,\sigma) = F(\sigma,\rho)$ and $F \in [0,1]$.
2. A continuous variable teleportation system cannot reach a 100% fidelity.

THANK YOU FOR YOUR LISTENING